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SCIENCE MADE VISUAL

5 Illustrated Science Study Sheets

See It. Understand It. Remember It.

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The same carbon atoms are reused again and again!

The Carbon Cycle

Carbon moves, life continues!

A healthier planet for brighter tomorrows!

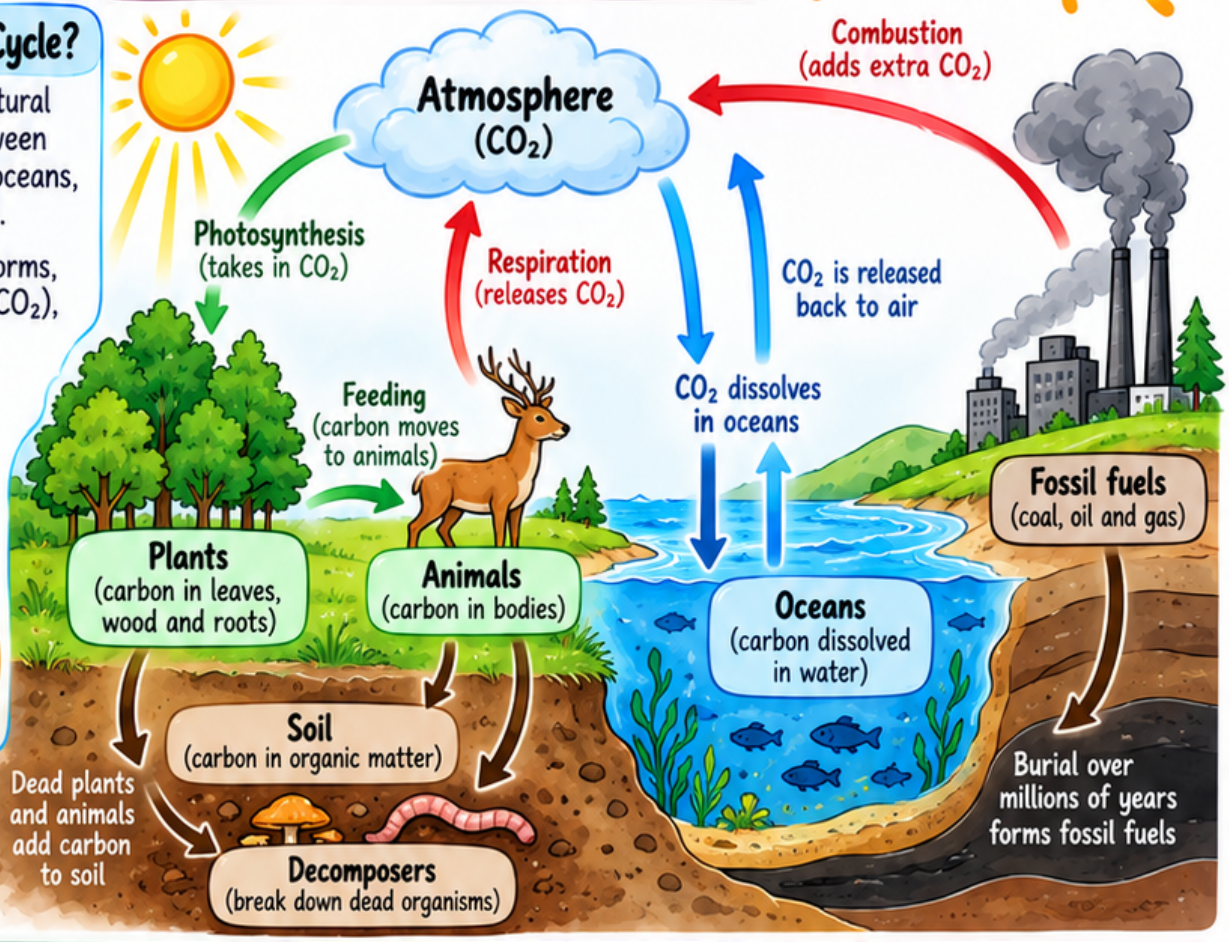
What is the Carbon Cycle?

The carbon cycle is the natural movement of carbon between Earth's air, living things, oceans, soil, rocks and fossil fuels.

Carbon is found in many forms, including carbon dioxide (CO_2), in living organisms and in rocks and fuels.

CO_2

The same carbon atoms move between different places!



Short-term Carbon Movement

Carbon moves quickly between air, plants, animals, soil and oceans, often in days, years or decades.



Long-term Carbon Storage

Some carbon is stored for hundreds, thousands or even millions of years in oceans, deep ocean sediments, soil and fossil fuels.



Over time, layers of sediment lock carbon away.

This long-term storage helps keep Earth's climate stable.

Human Impact

Burning fossil fuels releases carbon that has been stored underground for millions of years. This adds extra CO_2 to the atmosphere faster than natural processes can remove it.



Why the Carbon Cycle Matters

- Provides carbon for all living things (plants, animals and microbes).
- Helps regulate Earth's climate by keeping carbon in balance.
- Supports healthy oceans, soil and ecosystems.
- Without the carbon cycle, life as we know it could not exist!

A stable carbon cycle keeps our planet livable!



Reservoirs vs Fluxes

Carbon Reservoirs

Places that store carbon for a long time.



- Atmosphere
- Oceans
- Living things
- Soil
- Rocks and fossil fuels

Carbon Fluxes

Processes that move carbon from one place to another.



- Photosynthesis
- Respiration
- Decomposition
- Ocean exchange
- Sedimentation
- Combustion

Key Facts – Remember!

- Carbon cycles between air, living things, oceans, soil, rocks and fuels.
- Plants take in CO_2 and animals release CO_2.
- Oceans both absorb and release CO_2.
- Some carbon is stored for millions of years in rocks and fossil fuels.
- Burning fossil fuels adds extra CO_2 to the atmosphere.
- A healthy carbon cycle supports life and helps keep our climate stable.

The carbon we share today will be here tomorrow!

Chemistry is all around us!

Acids, Bases and the pH Scale

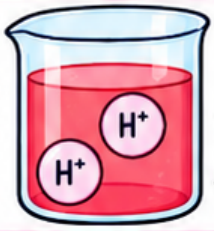
Everyday substances can be acidic, basic or neutral!

Different solutions, brighter understanding!

What are acids?

Acids are substances that produce hydrogen ions (H^+) in water. They have a pH less than 7, taste sour and can be corrosive.

Acids release H^+ ions in water.



More H^+ ions → lower pH

What are bases?

Bases are substances that produce hydroxide ions (OH^-) in water. They have a pH greater than 7, feel slippery and can be caustic.

Bases release OH^- ions in water.

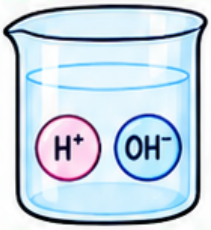


More OH^- ions → higher pH

Neutral solutions

Neutral solutions have equal amounts of H^+ and OH^- ions. They have a pH of 7 and are neither acidic nor basic.

In neutral water: same number of H^+ and OH^- ions.



pH 7 (neutral)

The pH Scale

The pH scale shows how acidic or basic a solution is. It ranges from 0 to 14.

More acidic (lower pH) ← → More basic (higher pH)

Neutral

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14



Battery acid (pH 0) · Lemon juice (pH 2) · Vinegar (pH 3) · Water (pH 7) · Baking soda (pH 8) · Soap (pH 10) · Bleach (pH 12)

Indicators

Indicators change colour depending on the pH of a solution. Universal indicator shows a range of colours, and litmus paper turns red in acids and blue in bases.

Universal indicator colours



Acidic (pH 0–6) · Neutral (pH 7) · Basic (pH 8–14)

Litmus paper



Turns red in acids · Turns blue in bases

Properties of acids and bases

Property	Acids	Bases
pH value	Less than 7	Greater than 7
Taste	Sour (never taste in the lab!)	Bitter (never taste in the lab!)
Feel	Can be corrosive	Feel slippery or soapy
Reaction with metals	Often react to produce hydrogen gas	Usually do not react with metals
Effect on indicators	Turn blue litmus red (universal indicator shows red/orange/yellow)	Turn red litmus blue (universal indicator shows blue/purple)

Safety notes



Wear safety goggles and a lab coat.



Avoid skin and eye contact with acids and bases.



If a spill happens, tell an adult and follow safety procedures.



Always wash your hands after handling chemicals.

Be safe in the lab – science is amazing!

Key facts to remember

1. Acids have pH less than 7.
2. Bases have pH greater than 7.
3. Neutral solutions have pH 7.
4. Indicators change colour with pH.
5. Many everyday substances are acids or bases.

Understanding pH helps us in everyday life and keeps us safe!

Small changes in pH make a big difference!

Science makes a brighter, safer world!

The Human Heart: Chambers, Valves and Blood Flow

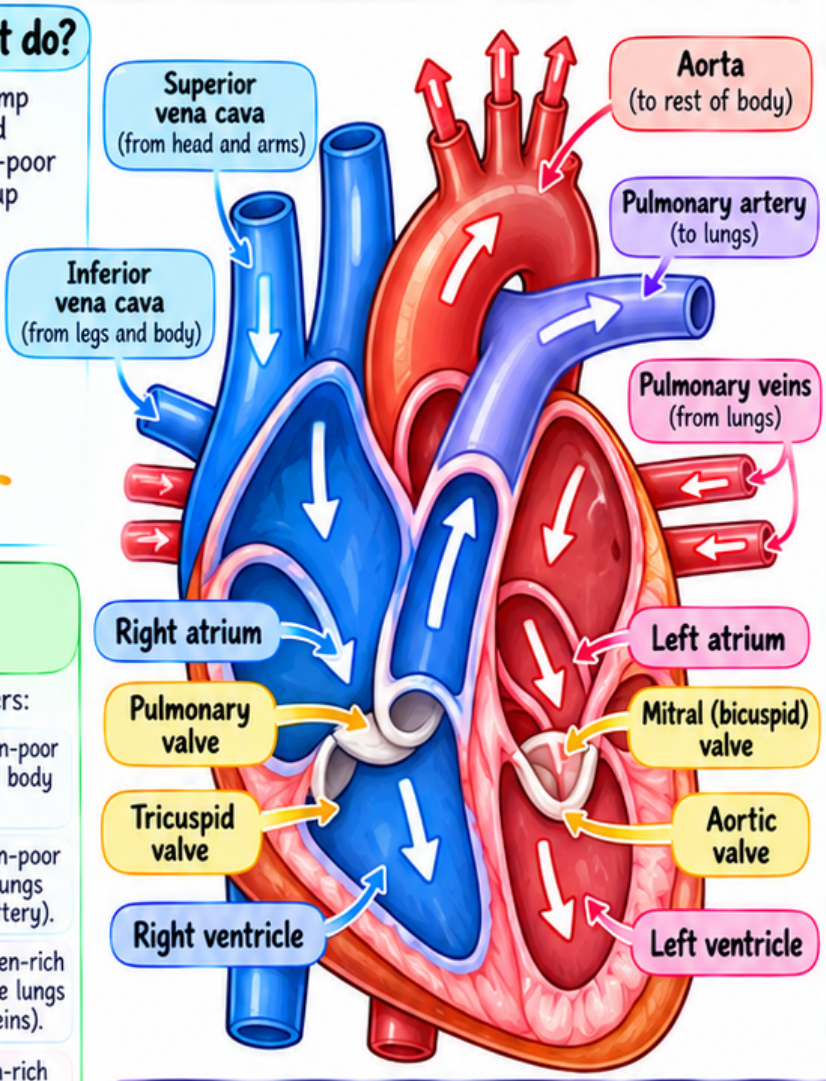
A powerful pump that keeps us alive!

Your heart beats about 100,000 times every day!

What does the heart do?

The heart is a muscular pump that circulates blood around your body. It sends oxygen-poor blood to the lungs to pick up oxygen, and then pumps oxygen-rich blood to every cell in your body.

Your heart beats about 70 times per minute, pumping around 5 litres of blood every minute!



The chambers of the heart

The heart has four chambers:

Right atrium	Receives oxygen-poor blood from the body (vena cava).
Right ventricle	Pumps oxygen-poor blood to the lungs (pulmonary artery).
Left atrium	Receives oxygen-rich blood from the lungs (pulmonary veins).
Left ventricle	Pumps oxygen-rich blood to the body (aorta).

The heart valves

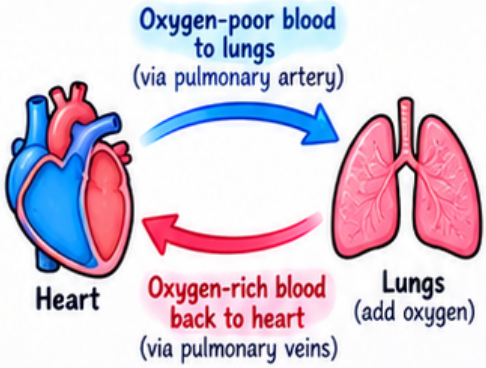
The heart has four valves that open and close to keep blood moving in the right direction. They prevent backflow of blood.

Tricuspid valve Between right atrium and right ventricle.	Stops blood flowing back into the right atrium.
Pulmonary valve Between right ventricle and pulmonary artery.	Stops blood flowing back into the right ventricle.
Mitral (bicuspid) valve Between left atrium and left ventricle.	Stops blood flowing back into the left atrium.
Aortic valve Between left ventricle and aorta.	Stops blood flowing back into the left ventricle.

Valves open as pressure pushes blood forward, and close to prevent backflow!

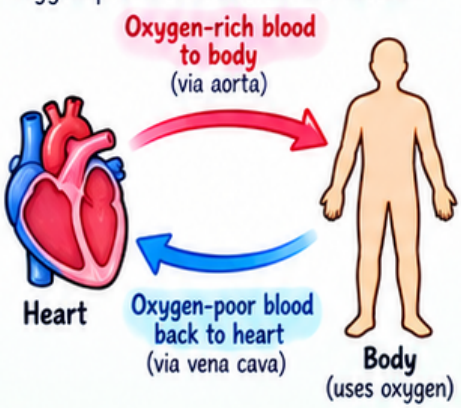
Pulmonary circulation (blood to the lungs)

Carries oxygen-poor blood from the heart to the lungs, where it picks up oxygen and releases carbon dioxide.



Systemic circulation (blood to the body)

Carries oxygen-rich blood from the heart to the rest of the body, and returns oxygen-poor blood back to the heart.



Key facts to remember

- The human heart has four chambers and four valves.
- The right side of the heart pumps oxygen-poor blood to the lungs (pulmonary circulation).
- The left side of the heart pumps oxygen-rich blood to the body (systemic circulation).
- The pulmonary artery carries oxygen-poor blood, and the pulmonary veins carry oxygen-rich blood (opposite to most veins!).
- Arteries carry blood away from the heart. Veins carry blood towards the heart.
- Your heart beats about **100,000** times every day, pumping around **7,000** litres of blood!

A healthy heart helps you live a longer, happier life!

A healthy heart powers a brighter tomorrow!

Take care of your heart!

Electric Current, Voltage and Resistance

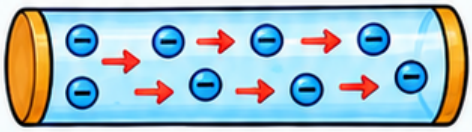
Electricity powers our world!

Same ideas. Brighter tomorrows!

Three key ideas that make electricity work!

Electric Current (I)

Electric current is the flow of electric charge (tiny particles called electrons) through a material, usually a wire.



Current tells us how many electrons pass a point each second.

Voltage (V)

Voltage is the electrical push that makes electrons move around a circuit. It is like a pressure difference between two points.



Higher voltage gives a stronger push for the electrons.

Resistance (R)

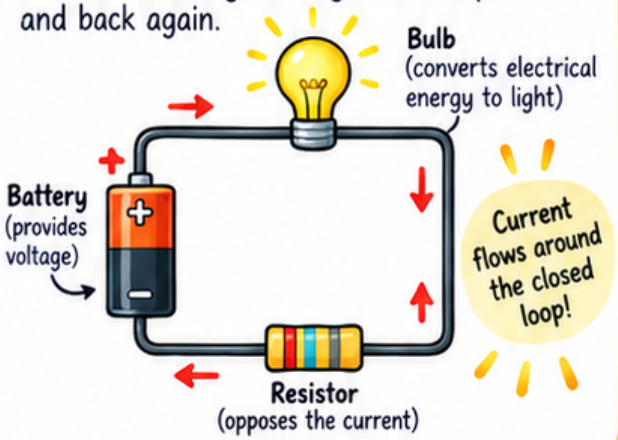
Resistance is the opposition to the flow of electric current. It makes it harder for electrons to move through a material.



Higher resistance reduces the flow of current.

A Simple Electric Circuit

A complete circuit lets current flow from the battery, through the components, and back again.



Ohm's Law

Voltage, current and resistance are related by **Ohm's law**:

$$V = I \times R$$

V = voltage (volts, V)
I = current (amperes, A)
R = resistance (ohms, Ω)

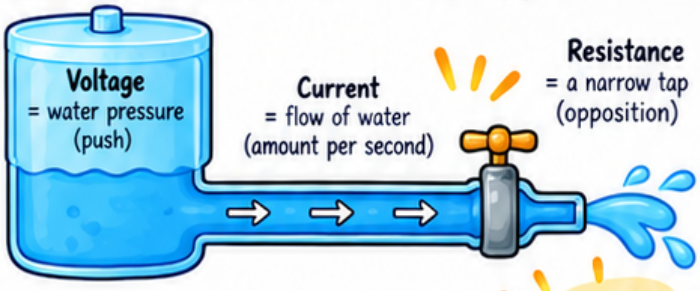
If you know two of these values, you can find the third using this formula!

Side by Side Comparison

Voltage (V)	• The electrical push • Measured in volts (V) • Supplied by a battery or power source
Current (I)	• The flow of charge • Measured in amperes (A) • Shows how many electrons pass each second
Resistance (R)	• The opposition to flow • Measured in ohms (Ω) • Depends on the material, length, thickness and temperature

The Water Analogy

Electricity is a bit like water flowing in pipes!



More pressure → more flow
Bigger resistance → less flow

Same idea, different stuff!

Real Examples

Higher voltage (thicker push)



Using a bigger battery increases the current (if resistance stays the same).

Double the voltage → double the current (same resistance).

Higher resistance (more opposition)



A larger resistor reduces the current (if voltage stays the same).

Double the resistance → half the current (same voltage).

Effect on current



More voltage makes the bulb brighter. More resistance makes it dimmer.

Current depends on both voltage and resistance!

Key Facts to Remember

- Electric current is the flow of electrons (A).
- **Voltage** is the electrical push that drives the current (V).
- **Resistance** is the opposition to the flow (Ω).
- They are linked by Ohm's law: $V = I \times R$.
- Higher voltage increases current (if resistance is the same).
- Higher resistance decreases current (if voltage is the same).

Small ideas make a big difference!

The Periodic Table of Elements

All the elements, in a clever pattern!

Same building blocks, bright possibilities!

What is an element?

An element is a pure substance made of only one type of **atom**. Each element has its own unique properties. There are **118** known elements.

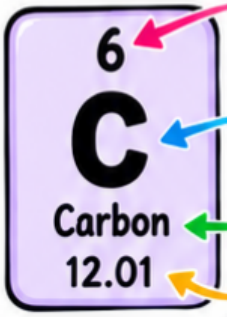


Gold (Au) | Carbon (C) | Oxygen (O)

Elements cannot be broken down into simpler substances by ordinary chemical methods.

An element tile

Each element on the periodic table is shown in a tile with three key pieces of information:



- **Atomic number** (tells us how many protons an atom has)
- **Symbol** (a short letter or two letters)
- **Name**
- **Relative atomic mass** (average mass of the atom)

How the table is arranged

The elements are arranged in order of **increasing atomic number** (from 1 to 118).

Atomic number increases →

1 | 2 | 3 | 4 | 5 | ... | 118

As you move from left to right, the atomic number increases by 1.

This arrangement shows a repeating pattern in the properties of the elements.

The Periodic Table

Elements are arranged in rows and columns with a repeating pattern of properties!

Groups go up and down

Groups (columns)

Periods (rows)

Each row is called a period. Properties change gradually as you move from left to right.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H																	He
2	Li	Be											B	C	N	O	F	Ne
3	Na	Mg											Al	Si	P	S	Cl	Ar
4	K	Ca											Ga	Ge	As	Se	Br	Kr
5	Rb	Sr											In	Sn	Sb	Te	I	Xe
6	Cs	Ba											Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra																

Transition metals (groups 3–12)

Lanthanides (rare earth metals)	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Actinides (radioactive metals)	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Groups (columns)

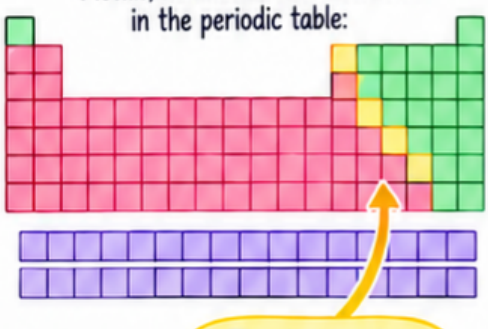
Vertical columns are called groups. Elements in the same group have similar properties because they have the same number of electrons in their outer shell.

Areas of the periodic table

Elements can be grouped into metals, nonmetals and metalloids. Each area has different properties.

- **Metals** — Most elements (left side and centre). Usually shiny, good conductors of heat and electricity, and solid at room temperature (except mercury).
- **Nonmetals** — Found on the right side (and hydrogen). Often gases or brittle solids. Poor conductors of heat and electricity.
- **Metalloids** — A small group along the "staircase" line. Have properties between metals and nonmetals.

Metals, nonmetals and metalloids in the periodic table:



The staircase line shows the metalloids.

Examples from each area

Metal: Iron (Fe)



- Shiny
- Good conductor
- Solid

Nonmetal: Oxygen (O)



- Gas
- Poor conductor
- Essential for life

Metalloid: Silicon (Si)



- Semiconductor
- Properties between metals and nonmetals
- Used in electronics

Key Facts to Remember

- There are 118 known elements.
- Elements are arranged by increasing atomic number.
- Rows are called periods.
- Columns are called groups.
- Elements can be metals, nonmetals or metalloids.
- Elements in the same group have similar properties.

Different elements. A brighter world!

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